Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/30/2023
PI Name:	Pathak, Siddhartha Ph.D.		
Project Title:	Understanding the Local Structure-Property Relationships of Solders in Terrestrial vs. Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Ames	**State:**	IA
Zip Code:	50011	**Congressional District:**	4
Comments:	NOTE: PI moved in fall 2020 to Iowa State University from University of Nevada, Reno.		
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-E: Use of the NASA Physical Sciences Informatics System – Appendix E
Start Date:	01/01/2020	**End Date:**	12/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Panda, Binayak	**Contact Phone:**	
Contact Email:	binayak.panda-1@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2022 per NSSC information (Ed, 1/15/23) NOTE: Principal Investigator (PI) Dr. Sid Pathak moved to Iowa State University in the fall of 2020 and remains as PI through a subcontract. Dr. Jeffrey LaCombe is now the administrative PI at University of Nevada, Reno (Ed., 11/23/2020)		
Key Personnel Changes/Previous PI:	Year 1 Report: The Principal Investigator (PI) Dr. Sid Pathak has moved from the University of Nevada, Reno (UNR) to Iowa State University (ISU), Ames. The primary PI at UNR has been changed to Dr. Jeffrey LaCombe. UNR sub-contracted the funds for this project to Dr. Pathak at ISU.		
COI Name (Institution):	Zaeem, Mohsen Asle Ph.D. (Colorado School of Mines)		
Grant/Contract No.:	80NSSC20K0223		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The ongoing project is an integrated experimental and computational effort to characterize both the microstructure and resultant micro-to-nano mechanical response of solders in terrestrial vs. microgravity environments, 1 g vs. ~1×10^(-5) g. Results from the In-Space Soldering Investigation (ISSI) experiments performed aboard the International Space Station (ISS) have shown that soldering in microgravity is expected to be considerably different than their ground-based counterparts due to Earth's natural convective flow and buoyancy effects being minimized in microgravity during melting and solidification. The ISSI data has demonstrated that a lack of buoyancy forces in microgravity can internally trap the flux created during soldering at interfaces, such as repair joints. Our current hypothesis is that such internal porosity can be detrimental to the desired strength of the joint, as well as its thermal and electrical conductivity; our study aims to develop a comprehensive understanding of these effects in the millimeter-sized solders using an array of nano-mechanical testing tools, such as indentation and focused ion beam (FIB) fabricated micro-pillar compression and microtensile testing. Results from this study will be instrumental in enhancing our fundamental understanding of the effects of surface tension driven convection phenomena during solidification processing operations such as brazing, soldering, and welding. Furthermore, the microgravity experiments represent the lowest gravity boundary condition. As such, these results could also be useful in predicting solidification behavior on other lower gravity environments (e.g., Moon or Mars). We had two graduate students working on a cutting-edge experimental-computational research project about soldering and solidification in microgravity conditions. One graduate student at Iowa State University (ISU), Ames is working on the experimental part, and another graduate student at Colorado School of Mines (Mines) is working on the computational part of this project. We had three undergraduate senior design groups who have also worked on this project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ongoing project is unique in that it combines both experimental and computational research on solder samples from the microgravity and ground-based ISSI experiments, along with a comparative study on fresh solder samples. The insights gained from this work can be extended to study the effects of fluid interfaces with surface tension variations that are also common for many spaceflight-enabling technologies (e.g., boiling, heat transfer, combustion, welding, brazing, or soldering). We envision that this project will contribute to developing an advanced understanding of multiple processes that typically rely on gravity-driven separations in the terrestrial environment, such as the removal of gas bubbles from a liquid, which must be accomplished by other means in microgravity environments. Furthermore, the microgravity experiments represent a lowest gravity boundary condition. As such, these results could also be useful in predicting solidification behavior on other lower gravity environments (e.g., Moon or Mars).
Task Progress:	The purpose of this research is to better understand solidification behavior of solder materials under microgravity and surface driven convection conditions. In reduced gravity environments, such as aboard the International Space Station, it has been observed that the solidification of solder is negatively affected. In a terrestrial environment, the primary mode of mass and heat transport is driven by natural convection. Natural convection results in improved mixing and thermal diffusion due to gravity driven buoyant forces. Reductions in gravity shift the primary mode of mixing to surface tension-dominated Marangoni convection. This shift results in physical changes in the solidified solder, including differences in morphology and porosity. These changes are expected to negatively alter the mechanical properties of the solder joint. Solder joint porosity is a common but undesirable feature naturally arising from the use of fluxes and is more insidious in soldering joints formed under low-gravity conditions. Commercial soldering processes require the use of fluxes to suppress oxidation and promote wetting, which is necessary for effective solder spreading and joint-gap filling during solder reflow. Under typical soldering conditions, rapid reflow temperatures result in flux volatilization and bubble formation. It has been established that porosity arising from flux volatilization, which is dispersed and expelled from the solder joint under terrestrial gravity, may become entrapped within the freezing solder material under microgravity conditions, given the absence of buoyancy-driven convection (causes reduced fluid motion). Such porosity is seen in conventional (Earth gravity) solder joints but is much more prevalent in solder joints produced under reduced-gravity conditions. As a result, any solder joints made under reduced gravity (such as in-orbit repair of electronic devices, tubing, and mechanical joints) are at a risk of having their electrical or mechanical performance substantially degraded by entrapped porosity, increasing the associated risks of subsequent device failures on astronauts and equipment in the unforgiving environment of space. In this work, using two microgravity solders from the ISSI study, one ISSI terrestrial solder along with a third set of freshly made terrestrial 40wt%Pb-60wt%Sn solder, we report results from the microstructural characterization and resultant micro-to-nano mechanical response of solders in terrestrial vs. microgravity environments, 1g vs. ~1×10-5g under different temperature conditions. Results from the In-Space Soldering Investigation (ISSI) experiments performed aboard the International Space Station (ISS) have shown that soldering in microgravity is expected to be considerably different than ground based counterparts due to Earth's natural convective flow and buoyancy effects being minimized in microgravity during melting and solidification. On Earth, tests showed the solder melting sagging, dropping, or running down the test wire due to gravity effect. During this process, the entrapped gases and water vapor evolving from the circuit board itself are transported to the joint surface and mostly eliminated while the joint is molten. In comparison, aboard the ISS the molten solder was observed to form an equilibrium "football" shape. The changes in geometric shape are believed to be due to the dominance of surface tension in microgravity. Shortly after equilibrium, the accrued flux was seen to spin around the sample's surface. Liquid flux droplets rotating on the sample surface gave indirect evidence of gravity independent thermocapillary flow, induced by internal temperature gradients within the molten solder ball. A flux bubble would migrate up a temperature gradient to the hottest spot, i.e., the components being joined. The trapped flux poses several concerns as it concentrates at sites like a repair joint, which can be detrimental to the desired strength as well as thermal and electrical conductivity from soldering. Also, the solder samples prepared aboard the ISS exhibited a considerable amount of internal porosity (about 13 times that of ISSI ground-based solders). The 3-D microstructural information from non-destructive micro-computed tomography (micro-CT) demonstrate the 3-D distribution of voids in the microgravity solder. The increased porosity is believed to be due to the lack of buoyant forces on flux and water vapor inherently present in the molten solder. Entrapped gases will remain within the molten pool of solder until the solidification is complete, resulting in higher amount of voids, and compromising the mechanical integrity and electrical properties of the joint. Our current results also show a distinct difference of the distribution of the Sn-rich and Pb-rich regions of the terrestrial vs. microgravity solders using electron-backscattered imaging, and a lower strength in the microgravity solders from nanoindentation testing. A further consideration is the substantial effect of aging on the solder microstructure and properties over the past 17 years. Our results from this aging study demonstrate how the microstructure and properties of solder joints change as a

	function of aging. In addition to our work at the Iowa State University, the non-destructive tomography experiments conducted at the Advanced Photon Source demonstrate the 3-D distribution of voids and Pb-rich dendrites in the 40Pb-60Sn solder. Our multiphase field model (MPF) for predicting solidification nano/microstructures in microgravity and terrestrial conditions include three stages: 1) MPF model development that can properly predict the effect of composition (concentration of alloying elements) and solid-liquid interface property of the soldering material on solidification microstructure formation, 2) fully coupling to Navier-Stokes equations to investigate the effects of buoyancy and melt flow on formation of solidification microstructures and voids, and 3) coupling the thermo-elasto-viscoplastic governing equations to the MPF model, enabling prediction of stress/stain fields generated in soldering process, which also facilitates prediction of solidification shrinkage. This project has resulted in three senior design projects: 1) Senior Design Project 2020: Solder Joint Testing System, 2) Soldering Phase 1: Experimental Design for Investigating Microstructures and Mechanical Properties of Solder Samples, and 3) Pb-Sn Solders for In-Space Applications: Studying the Effect of Aging on Structure and Properties. This project has provided eight senior undergraduate students (3 senior design teams) and two sophomore undergraduate students with an opportunity to work on in-space soldering and to get hands-on experience on the lab instruments. In addition, this project has resulted in nine presentations at various conferences (e.g., the International Space Station Research and Development Conference/ISSRDC; Meeting of the American Society for Gravitational and Space Research/ASGSR) and two awards (Georgetown Scientific Research Journal cover awards and ASGSR travel award).
Bibliography Type:	Description: (Last Updated: 04/22/2025)